Tuple Worksheet Questions

Class XI Computer Science (083) — Topic: Tuples in Python

Instructions: Write the answers in the space provided or on a separate sheet as directed by your teacher.

Q1. What do you mean by Tuple in Python?

Q2. Write a statement to create an empty tuple named `T1`.

Q3. Write a statement to create a tuple `T1` containing first five even numbers.

Q4. Write the code to convert the given list `L1` to tuple.
`L1 = [1, 2, 3, 4, 5]`

Q5. Write a statement to create tuple `T1` with single element.

Q6. Write the output of the following code:

```
T1 = (4)
print(type(T1))
```

Q7. Write the output of the following code:

```
T1 = 4, 5, 6
print(type(T1))
```

Q8. Write the output of the following code:

```
T1 = (1, 2, 3, 4, 5, 6, 7, 8)
print(T1)
print(T1 * 2)
print(T1 + T1)
print(len(T1) * 2)
```

Q9. Write a statement to create tuple `T1` with the following data:
`1, 4, 'CS', 'IP', 5`

Q10. Write the output of the following code:

```
T1 = (1, 2, 3, 4, 5, 6, 7, 8)
print(T1[0])
print(T1[-1])
print(T1[2 + 3])
print(T1[4 - 1])
print(T1[7 % 2])
```

Q11. Write the output of the following:

```
T1 = (1, 2, 3, 4, 5, 6, 7, 8)
print(T1[:])
print(T1[3:])
print(T1[:4])
print(T1[-2:-5])
```

Q12. Write the output of the following:

```
T1 = (1, 2, 3, 4, 5, 6, 7, 8)
print(T1[1::2])
print(T1[-1:-5:-2])
print(T1[::-1])
print(T1[:7:2])
```

Q13. Which error is returned by the following code:

```
T1 = (1, 2, 3, 4, 5, 6, 7, 8)
print(T1[15])
```

Q14. Write two differences between Tuple and List.

Q15. Write a statement in Python to concatenate the given tuples.

```
T1 = (1, 2, 3)
T2 = (5, 6, 7)
```

Q16. Write the output of the following code:

```
T1 = (45, 67, 98)
T1 = T1 + (1, 2, 3)
print(T1)
```

Q17. Write the output of the following code:

```
T1 = (45, 67, 98)
T1 = T1 * 3
print(T1)
```

Q18. Write the output of the following code:

```
T1 = (45, 67, 98)
T2 = (45, 67, 98)
print(T1 in T2)
print(45 in T2)
print(45 in T1)
print(T1 + T2)
```

Q19. Explain the following functions in reference to Tuple in Python:

a. `len()`

b. `count()`

Q20. Write a program to accept five numbers from the user and store it in a tuple `'T1'`.

Q21. Write a program to accept five fruit names from the user and store it in a tuple `'F1'`.

Q22. Write a program to store `'n'` number of sports in a tuple `'S1'`. (Accept `'n'` from the user)

Q23. Consider the following tuple and write the code for the following statements:

```
T1 = (12, 3, 45, 'Hockey', 'Anil', ('a', 'b'))
```

a. Display the first element of `'T1'`

b. Display the last element of `'T1'`

c. Display `'T1'` in reverse order.

d. Display `'Anil'` from tuple `'T1'`

e. Display `'b'` from tuple `'T1'`

Q24. Write the output of the following:

```
for i in tuple("SHIMLA"):
    print(i + i)
```

Q25. Write the output of the following:

```
>>> 7 in tuple('123456789')
```

Q26. Write the output of the following:

```
T1 = (23, 32, 4, 5, 2, 12, 23, 7, 9, 10, 23)
print(len(T1), *T1[-1])
print(T1[T1.count(23) + len(T1) - 5])
print(T1.count(T1[6]))
print(T1.count(T1[-1]))
```

Q27. Write the output of the following:

```
T1 = (23, 32, 4, 5, 2, 12, 23, 7, 9, 10, 23)
print(max(T1))
print(min(T1))
```

Q28. Write the output of the following:

```
T1 = ('Hockey', 'Cricket', 'Football')
print(max(T1))
print(min(T1))
```

Q29. Write the output of the following:

```
t1 = ('Raman', 'Ram', 'Ramaiya')
print(max(t1))
print(min(t1))
```

Q30. Write the output of the following:

```
T1 = (23, 32, 4, 5, 2, 12, 23, 7, 9, 10, 23)
print(sorted(T1))
print(sorted(T1[2:7]))
print(T1.index(23))
print(T1.index(23, 3, 9))
```

Q31. Write the output of the following:

```
T1 = (23, 32, 4, 5, 2, 12, 23, 7, 9, 10, 23)
print(sum(T1) + min(T1))
```

(Note: Original may have had a typo in map(); corrected for sense)

Q32. Write a program to accept three numbers from the user and insert it at the end of given Tuple T1.
`T1 = (23, 32, 4, 5, 2, 12, 23, 7, 9, 10, 23)`

Q33. What type of error is returned by following statement:

```
>>> (a, b, c, d) = (5, 6, 8)
```

Q34. Write the output of the following code:

```
>>> (name, rollNo, subject) = ('Anil', 9, 'CS')
>>> name
```

Q35. Predict the output:

```
T = (1, 2, 3)
del T
print(T)
```

Q36. What do you mean by Nested Tuple? Give one example.

Q37. Write a statement to print 30 from the given tuple.
`a = ("Seventy", [1, 2, 3], (20, 30, 40), "Eighty")`

Q38. Write a statement to unpack the following tuple into 3 variables.
`(60, 70, 80)`

Q39. Write a program to swap the values of given tuples.
`T1 = ('A', 'B')`
`T2 = (34, 65)`

Q40. Write the output of the following code:

```
T1 = ('A', ['B', 'D'], 'C')
T1[1][1] = 'C'
print(T1)
```

Q41. Write a program to print the frequency of a number accepted from the user in given tuple:
`T1 = (12, 17, 18, 25, 19, 12, 18, 5)`

Q42. Write a program in Python to concatenate all the characters of given tuple.
`T1 = ('B', 'O', 'O', 'K')`
Expected OUTPUT : BOOK

Q43. Write a program to remove a number (accepted from the user) from the given tuple
`T1 = (12, 15, 18, 21, 24, 27, 30)`
SAMPLE EXECUTION
Enter element to remove : 21
Tuple after removing element is : (12, 15, 18, 24, 27, 30)

Q44. What do you mean by Unpacking Tuple? Give example.

Q45. What do you mean by Packing Tuple? Give Example.

Q46. Write one similarity between Tuples and String.

Q47. Write the output of the following:

```
T1 = (2, 'Apple', 6)
print(max(T1))
```

Q48. Consider the given tuple and write the output of given statements:
`T1 = (1, 23, 4, 5, 'A', ['C', 'D'], (23, 45), 45)`

a. `print(len(T1))`

b. `print(T1.index(45))`

c. `print(T1.count(45))`

d. `print(T1[5][0] - 2)`

e. `print(T1[:5])`

f. `print(T1[5:])`

g. `print(T1.index("A"))`

h. `print(T1[-1:-7:-2])`

i. `print(T1[4] + T1[5])`

j. `print(max(T1))`

Q49. Write the length of following tuple:
`T1 = (1, (2,3), ((34,56),45,67), 39)`

Q50. Write a program to store the detail (Roll number, Name, Section) of three students in a tuple. Accept all data from the user.
SAMPLE OUTPUT :
`((1, 'Amita', 'B'), (2, 'Sunita', 'C'), (3, 'Daya', 'A'))`

Q51. Write a program to accept roll number from user and display it's detail from given tuple
`Det = ((1, 'Amita', 'B'), (2, 'Sunita', 'C'), (3, 'Daya', 'A'))`

Q52. Write a program to accept five numbers from the user and store these numbers in a tuple. Display the following from tuple:

a. Largest number

b. Smallest number

c. Sum of all numbers

d. Average of all numbers

Q53. Write a program in Python to print the second largest element in tuple given below:
`T1 = (23, 45, 87, 45, 67, 43, 23, 12)`

Q54. Write a program in Python to display the sum of all even numbers and odd numbers separately from the tuple given below:
`T1 = (24, 45, 87, 46, 67, 44, 23, 12)`

Q55. Write the output of the following:

```
T1 = ("Amit", "Sumit")
T2 = "Sumit", "Amit"
T3 = "Amit", "Sumit"
print(T1 == T2)
print(T1 == T3)
```

Q56. Write a program to accept a number from the user and store first 10 multiples of number in a tuple.
For example:
Enter any number : 4
Tuple = (4, 8, 12, 16, 20, 24, 28, 32, 36, 40)

Q57. Write a program to accept a string from the user and convert it into tuple.
For example:
Enter any String : tuple
Tuple = ('t', 'u', 'p', 'l', 'e')

Q58. Write a program to display the sum of all the numbers in a given tuple:
`T1 = (1, 2, 3, 'A', 'B', 4, 'Sun')`

Q59. Write a program to find the sum of marks of all students stored in the given tuple.
`T1 = (("Suman", 75), ("Glory", 35), ("Ravi", 50))`
Expected Output is : 160

Q60. Write the output of the following:

```
T1 = (1, (2, 'a'), (3, 'b', 'c'), (7, 'd', 'e', 'f'))
print(len(T1))
print(T1[1][0])
print(7 in T1[3])
print(T1.index(1))
print(len(T1[3]))
print(T1[T1[2][0]])
```

Q61. Write the output of the following:

```
T1 = (1, (2, 'a', (3, 'b', 'c', (7, 'd', 'e', 'f'))))
print(len(T1))
print(T1[1][0])
print(7 in T1[1])
print(T1.index(1))
print(len(T1[1]))
print(T1[T1[0]])
```

Q62. Write the output of the following:

```
T1 = (1, (2, 'a') * 2)
print(T1)
print(len(T1))
print(T1[1][1])
print(len(T1[1]))
```

Q63. Write the output of the following:

```
T1 = (1, (2, 'a')) * 2
print(T1)
print(len(T1))
print(T1[1][1])
print(len(T1[1]))
print(T1.count(1))
print(T1.index(1))
```

Q64. Write the output of the following:

```
T1 = ('a') * 3
T2 = ('a',) * 3
print(T1)
print(type(T1))
print(len(T1))
print(T2)
print(type(T2))
print(len(T2))
```

Q65. Write the output of the following:

```
T1 = "Spider Man:No Way Home"
T2 = tuple(T1.split())
print(T1)
print(type(T1))
print(len(T1))
print(T2)
print(T2[0][0] + T2[1][0])
print(len(T2))
```

Q66. What type of error is returned by following code:

```
T1 = (1, 2, 3, 4)
T2 = (4, 5, 6)
print(T1 - T2)
```

Q67. Write the output of the following:

```
T1 = (1, 2, 3, 4)
print(T1 * 3)
print(T1 * (3))
print((T1) * 3)
print(T1 + (3,))
```

Q68. Write the output of the following:

```
T1 = (30, 20, 30)
print(T1, T1, T1)
print(T1 * 3)
```

Q69. Write the output of the following:

```
a = 30
b = (30)
c = (30,)
print(b)
print(type(b))
print(c)
print(type(c))
```

Q70. Write a program to accept a number from the user and create a tuple containing all factors of the number.
Sample Output:
Enter any Number : 9
Tuple is : (1, 3, 9)

Q71. Write a program to arrange all the sub-tuples of a tuple in increasing order and store them in another tuple.
For example:
`T1 = ((1, 2, 3), (45, 23), (98, 34, 67))` *# Original Tuple*
`T2 = ((1, 2, 3), (23, 45), (34, 67, 98))` *# New Tuple*

Q72. Write a program to store the length of all sub-tuples of a tuple into a new tuple.
For example:
`T1 = ((1, 2, 3), (45, 23), (98, 34, 67), (34, 56, 78, 9))` *# Original*
`T2 = (3, 2, 3, 4)` *# New Tuple*

Q73. Write a program to print square of all odd numbers of a tuple.
`T1 = (12, 13, 22, 7, 9, 66, 4)`
Expected Output : 169, 49, 81

Q74. Write a program to print the common elements of given tuples:
`T1 = (1, 2, 3, 4, 5, 6)`
`T2 = (3, 5, 7, 9, 11)`
Expected Output : 3, 5

Q75. Write a program to find the union (common elements included once) of given tuples:
`t1 = (1, 2, 3, 4, 5, 6)`
`t2 = (3, 5, 7, 9, 11)`
Expected Output : 1, 2, 3, 4, 5, 6, 7, 9, 11

Q76. Write a program to store the sum of all sub-tuples of given tuple into a new tuple.
For example:
`T1 = ((1, 2, 3), (45, 23), (98, 34, 67), (34, 56, 78, 9))` *#Original*
`T2 = (6, 68, 199, 177)` *# New Tuple*

Q77. Write a program to store the maximum value of all sub-tuples of given tuple into a new tuple.
For example:
```
T1 = ((1, 2, 3), (45, 23), (98, 34, 67), (34, 56, 78, 9), ("A", "B", "c"))
T2 = (3, 45, 98, 78, 'c') # New Tuple (note: string max)
```

Q78. Write a program to display the longest sub-tuple from the given tuple.
```
T1 = ((1, 2, 3), (45, 23), (98, 34, 67), (34, 56, 78, 9))
T2 = (34, 56, 78, 9) # New Tuple
```

Q79. Write a program to display names starting from vowel from the given tuple.
```
T1 = ("Amit", "Ram", "Esha", "Harry")
```

Q80. Write a program to print the length of all names in a given tuple.
```
T1 = ("Amit", "Ram", "Esha", "Harry")
```
Expected Output : 4, 3, 4, 5

Q81. What is the purpose of following operators in reference to tuples.

a. `+`

b. `*`

c. `in`

Q82. Match the following:

Functions	Description	Match
`max()`	Returns the length of the tuple.	(3)
`min()`	Returns the index value of element in tuple.	(4)
`count()`	Returns the largest element in tuple.	(1)
`index()`	Returns the smallest element in tuple.	(2)
`len()`	Returns the frequency of an element in tuple.	(5)

(Match the numbers correctly in your answer)

Q83. Write the output of the following:

```
>>> S = 1, (2,3,4,5), 51, (61,71)
>>> len(S)
>>> S[2:3]
>>> S[-1][0]
>>> type(S)
```

Q84. Predict the output:

```
T = (1, 2, 4, 3, 8, 9)
[T[i] for i in range(0, len(T), 2)]
```

Q85. What will be the output of the following code snippet?

```
init_tuple = ()
print(init_tuple.__len__())
```

— End of Questions —